



# **Rajshahi University of Engineering & Technology**

## **Rajshahi-6204, Bangladesh**

Department of Electrical & Computer Engineering (ECE)

Course No.: ECE 4000

### **Title : CRISISCLIP-X: EXPLAINABLE CLIP-BASED MULTIMODAL DISASTER CLASSIFICATION**

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# Outline of This Presentation

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## Context

- ❑ Disaster-related social media posts often contain both images and text.[1][2]
- ❑ These posts may provide useful information about damage, affected people, rescue activities, and urgent needs.
- ❑ However, many posts are noisy, duplicated, unclear, or irrelevant.[3]
- ❑ Manual screening becomes slow and difficult during emergency situations.

## Proposed Idea

- ❑ CRISISCLIP-X is an explainable CLIP-based multimodal framework for disaster classification.

### Input

Image + Text

### Tasks

Informativeness  
Humanitarian  
Severity

### Purpose

Support faster  
human review

**Goal: to classify disaster-related social-media posts more accurately and support structured emergency information screening.**

- ❑ During disasters, social media contains large volumes of image-text posts.
- ❑ Some posts are useful for emergency response, but many are irrelevant, noisy, duplicated, or unclear.
- ❑ Manual screening is slow and difficult during time-critical situations.[4]
- ❑ A multimodal AI system can help prioritize useful information for human review and situational awareness.



**Need: fast, structured, explainable disaster information management**

## Paper 1: Analysis of Social Media Data using Multimodal Deep Learning for Disaster Response[13]

### Method & Result

- ❑ Used CrisisMMD[8] image-text pairs for informativeness and humanitarian classification.
- ❑ Text branch: 5-layer CNN with word2vec; image branch: VGG16.
- ❑ Early feature fusion improved F1 to 84.2% for informativeness and 78.3% for humanitarian classification.

### Limitations

- ❑ Static word2vec and shallow CNN have limited contextual understanding.
- ❑ Simple concatenation does not model strong cross-modal interaction.
- ❑ Limited XAI support and weak handling of minority humanitarian classes.

**Relevance to this thesis: validated multimodal disaster learning and motivated stronger CLIP-based feature learning, imbalance handling, and XAI.**

# Literature Review (cont'd)

## Paper 2 : Analysis of Multimodal Social Media Data Utilizing ViT Base 16 and GPT-2 for Disaster Response[9]

### Method & Result

- ❑ Used ViT-B/16 for images and GPT-2 for text on CrisisMMD[8] informativeness classification.
- ❑ Combined unimodal predictions using Random Forest.
- ❑ Final multimodal model achieved 84.66% accuracy and ROC-AUC of 0.87.

### Limitations

- ❑ Fusion was decision-level, so rich hidden image-text interaction was lost.
- ❑ Only binary informativeness was tested; humanitarian and severity tasks were not evaluated.
- ❑ No XAI and no comparison with CLIP-style joint vision-language models.

**Relevance to this thesis: motivated feature-level multimodal fusion and direct CLIP-based image-text representation.**

## Paper 3 : Crisis-DIAS: Towards Multimodal Damage Analysis — Deployment, Challenges and Assessment[11]

### Method & Result

- ❑ Proposed a gated three-stage pipeline for informativeness, infrastructure damage, and severity.
- ❑ Used RCNN/FastText for text and Inception-v3 for images with attention fusion.
- ❑ Reported high F1 scores and real-time processing of about 100 tweets/min.

### Limitations

- ❑ Sequential gates can cause error propagation from earlier stages.
- ❑ Severity analysis mainly focused on infrastructural or utility damage.
- ❑ Limited XAI, missing-modality robustness, and cross-dataset validation.

**Relevance to this thesis: inspired sequential disaster screening, but this thesis adds CLIP, XAI, MEDIC evaluation, and parameter-efficient adaptation.**

## Observed Gaps in Prior Studies

- ❑ Many earlier models relied on CNNs, shallow text encoders, or simple multimodal fusion.[5]
- ❑ Several studies focused on only one task, especially informativeness, rather than a full disaster-analysis pipeline.
- ❑ Limited attention was given to explainable AI for interpreting both image and text predictions.
- ❑ Cross-dataset validation and practical deployment were often weak or missing.
- ❑ Minority classes and class imbalance remained difficult, especially for humanitarian categories and damage severity.

## How This Thesis Responds

- ❑ Uses CLIP ViT-B/32 as a stronger multimodal vision-language backbone.
- ❑ Covers three tasks together: informativeness, humanitarian category, and damage severity.
- ❑ Compares zero-shot prompting, adapter tuning, LoRA, and partial fine-tuning.[6]
- ❑ Integrates explainability through saliency maps, token attribution, and Gradient  $\times$  Input.
- ❑ Evaluates generalisation on MEDIC and demonstrates deployment through a Streamlit application.

*Based on the selected studies: Ofli et al. (2020), Urse et al., Gite et al. (2025), and Agarwal et al. (Crisis-DIAS, 2019)[11],[9],[13].*

**Main gap: prior work did not combine a strong CLIP-based backbone, parameter-efficient adaptation, XAI, cross-dataset validation, and practical deployment in one unified framework.**

# Objectives

01

## **Develop a Multimodal Framework:**

CLIP-based image-text disaster classification.

02

## **Perform Multi-task Analysis:**

Informativeness, humanitarian category, and severity.

03

## **Compare Adaptation Methods:**

Zero-shot, Adapter, LoRA, and partial fine-tuning.

04

## **Improve Reliability & XAI :**

Class imbalance handling and explainable predictions.

05

**Validate & Deploy :** CrisisMMD[8]/MEDIC evaluation and Streamlit prototype.

**Research Goal: Efficient + Explainable + Deployable Multimodal AI for Disaster Information Classification**

**Two complementary datasets were used to develop and evaluate the proposed multimodal disaster-classification framework.**

## **CrisisMMD[8]**

### **Primary dataset for model development**

Multimodal Twitter dataset containing disaster images and tweet text.

Collected from seven major natural disasters.

Supports informativeness, humanitarian-category, and damage-severity tasks.

Official splits support structured model training and evaluation.

## **MEDIC[15]**

### **Secondary dataset for generalisation testing**

Multimodal Dataset for Emergency and Disaster Image Classification.

Contains labels for informativeness, humanitarian category, damage severity, and disaster type.

Used for out-of-distribution and cross-dataset evaluation.

Helps test robustness beyond CrisisMMD.

**Dataset role: CrisisMMD builds the main models; MEDIC checks cross-dataset robustness.**

## Events Covered

| Crisis Event         | Collection Period   |
|----------------------|---------------------|
| Hurricane Irma       | Sep 6–21, 2017      |
| Hurricane Harvey     | Aug 25–Sep 20, 2017 |
| Hurricane Maria      | Sep 20–Nov 13, 2017 |
| California Wildfires | Oct 10–27, 2017     |
| Mexico Earthquake    | Sep 20–Oct 6, 2017  |
| Iraq-Iran Earthquake | Nov 13–19, 2017     |
| Sri Lanka Floods     | May 31–Jul 3, 2017  |

## Annotation Tasks

1

### Task 1

### Informativeness

Informative / Not Informative

2

### Task 2

### Humanitarian

Five aligned categories

3

### Task 3

### Damage Severity

Severe / Mild / Little-or-no damage

**CrisisMMD v2.0 multimodal baseline:**

**16,058 tweet texts and 18,082 images after cleaning**

**CrisisMMD v2.0 includes task-specific distributions and requires label harmonisation for multimodal learning.**

| Informativeness | Image  |
|-----------------|--------|
| Informative     | 9,374  |
| Not Informative | 8,708  |
| Total           | 18,082 |

| Humanitarian          | Image  |
|-----------------------|--------|
| Affected Individuals  | 562    |
| Infrastructure Damage | 3,624  |
| Not Humanitarian      | 8,708  |
| Other Relevant Info   | 2,529  |
| Rescue / Donation     | 2,231  |
| Total                 | 18,082 |

| Damage Severity    | Image |
|--------------------|-------|
| Little / No Damage | 475   |
| Mild Damage        | 839   |
| Severe Damage      | 2,212 |
| Total              | 3,526 |

**alignment decision:**

**Semantically close minority categories were merged: injured/dead + missing/found → Affected Individuals; vehicle damage → Infrastructure and Utility Damage.**

*Selected file: task\_informative\_text\_img\_agreed\_lab\_dev.tsv*

**Split**  
Development (Dev)

**Dataset Shape**  
1,573 rows × 9 columns

**Task**  
Informativeness  
(Text + Image)

## Preview Summary

The file was loaded successfully using pandas with TSV format.

This split contains multimodal disaster samples with tweet text and image references.

The preview confirms 9 structured fields for analysis and model input.

First 5 rows were displayed during exploratory data inspection.

## Columns

- event\_name
- tweet\_id
- image\_id
- tweet\_text
- image
- label
- label\_text
- label\_image
- label\_text\_image

**This preview slide summarises the CrisisMMD v2.0 development split used for informativeness classification.**

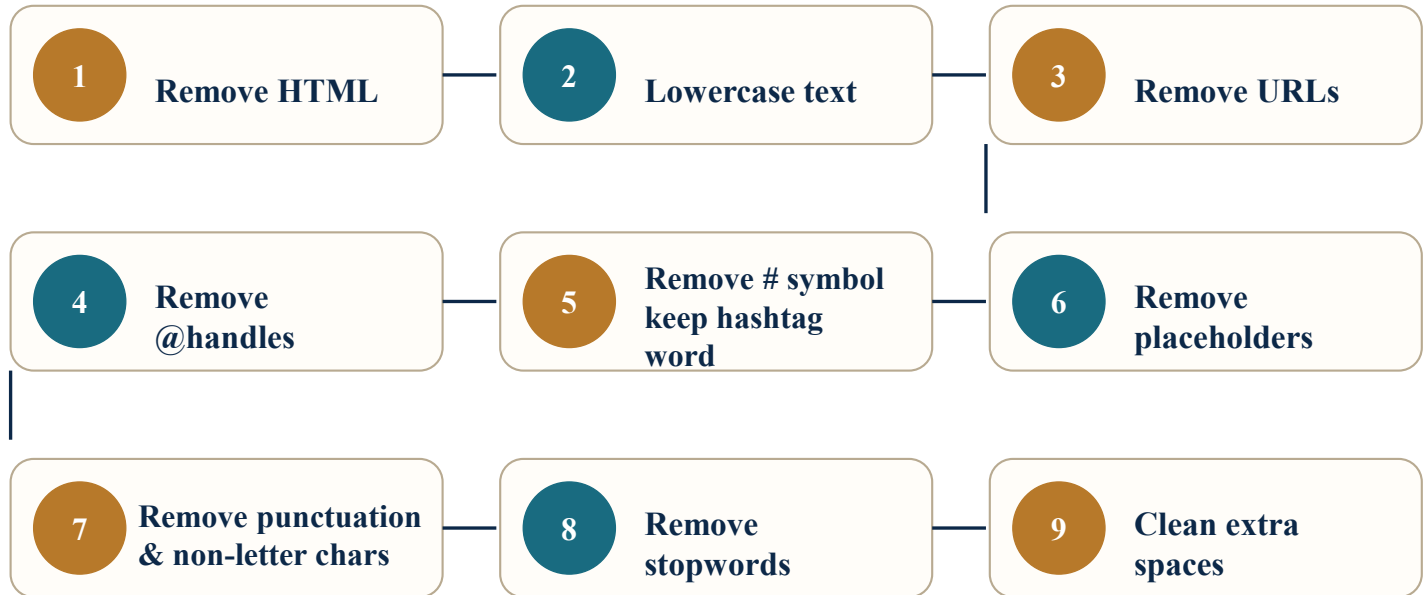
*Text preprocessing steps followed in the implemented code*

## Input Preparation

Text was converted to string format.

HTML was removed using BeautifulSoup.

The text was converted to lowercase. URLs, Twitter handles, placeholders, punctuation, and non-letter characters were removed.[7]



**Output: cleaned disaster-related tweet/caption text ready for CLIP tokenisation and model input.**

Note: Hashtag words were preserved after removing the “#” symbol; stopwords were removed using NLTK English stopwords.

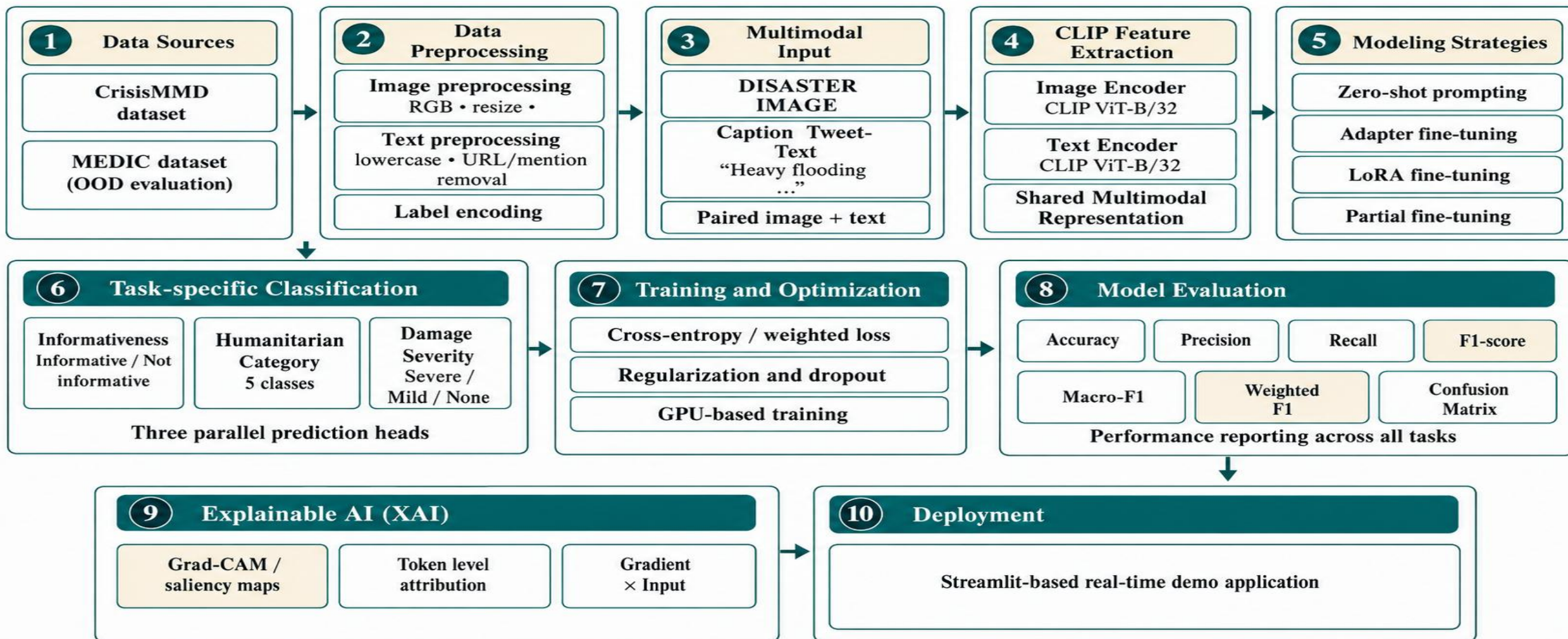


Figure : Proposed Methodology

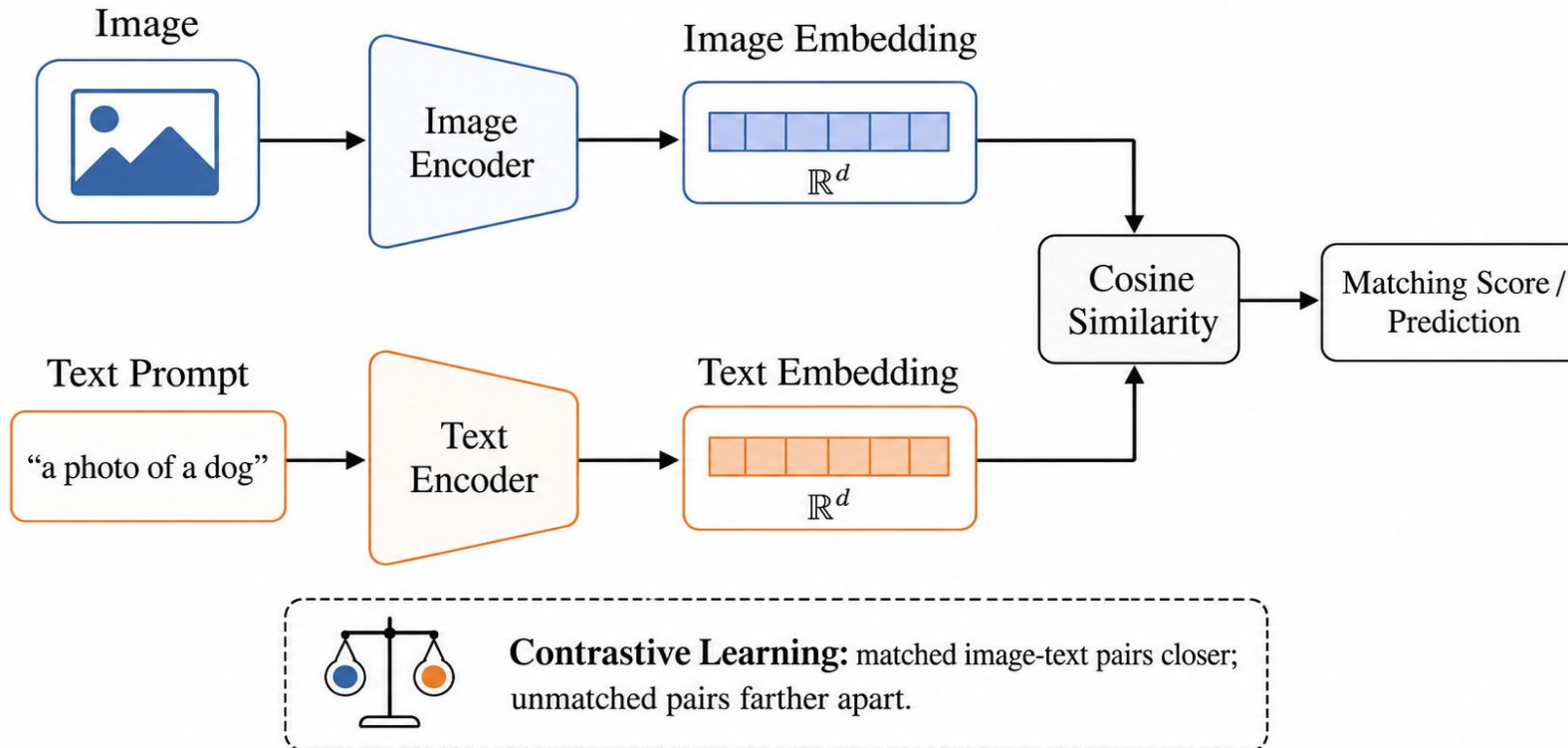


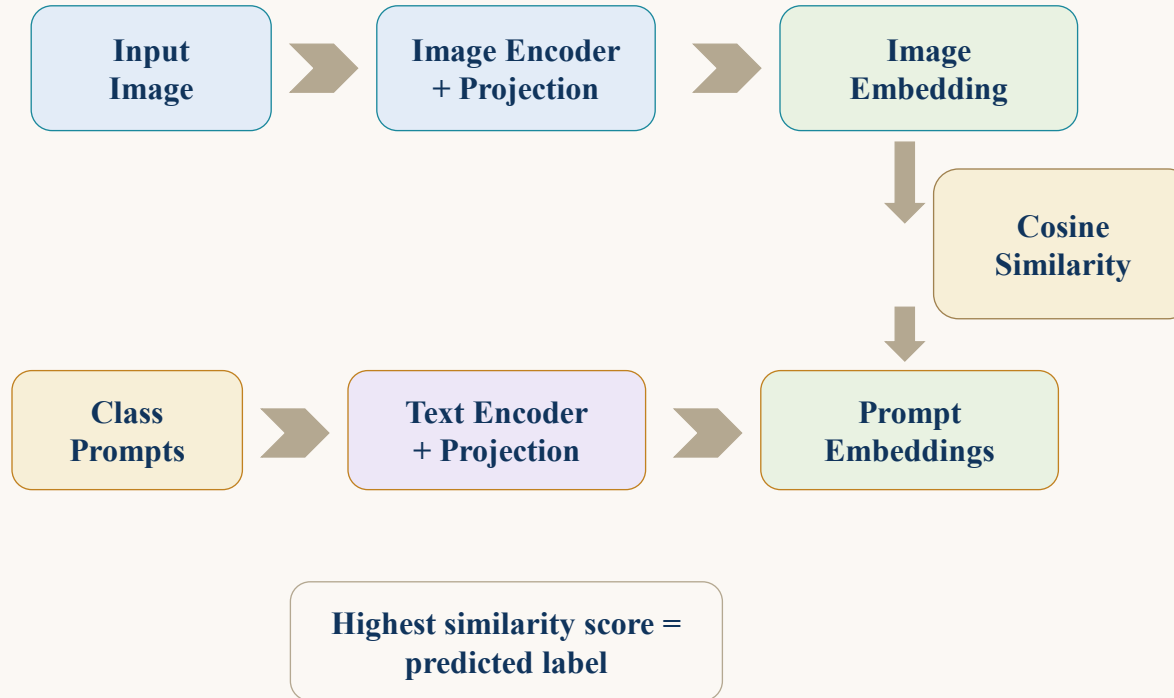
Figure : OpenAI CLIP Model Architecture

## Why Choose CLIP?

Processes both images and text through separate encoders.  
 Projects both modalities into a shared semantic embedding space.  
 Supports zero-shot classification using text prompts.  
 Provides a strong pretrained vision-language backbone for disaster posts.  
 Well suited for multimodal social-media analysis where image and caption must be interpreted

**Chosen as the shared backbone for all task variants.**

## Architecture Flow



## Why Choose Zero-Shot?

Requires no task-specific training.

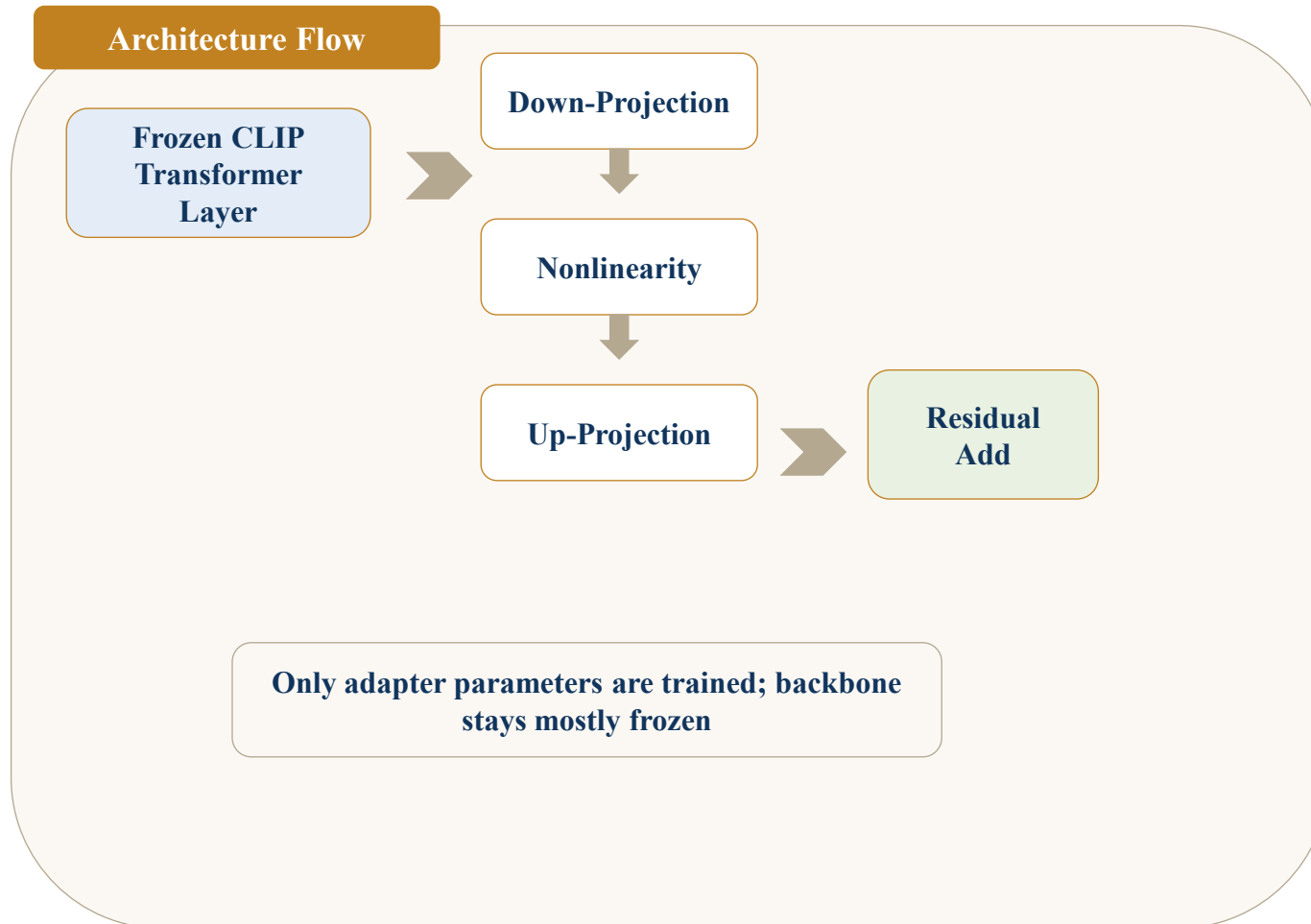
Uses only text prompts to define class labels.

Provides a clean baseline for comparison with fine-tuned models.

Computationally simple and easy to implement.

Helpful for checking whether pretrained CLIP knowledge already captures disaster concepts.

# Adapter-based fine-tuning



## Why Choose Adapters?

Insert small bottleneck modules inside transformer layers.

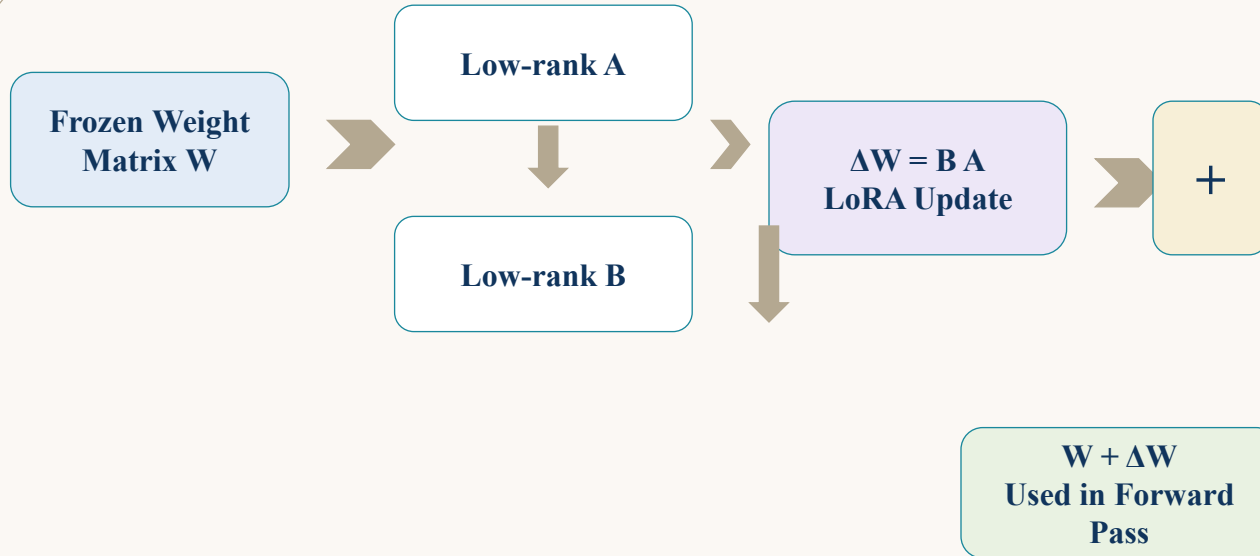
Train far fewer parameters than full fine-tuning.

Preserve most pretrained CLIP knowledge while learning task-specific transformations.

Effective when training data are limited or classes are imbalanced.

In this thesis, adapters produced the best humanitarian-classification performance.

## Architecture Flow



LoRA is applied to attention projections such as q\_proj, k\_proj, v\_proj, and out\_proj

## Why Choose LoRA?

Updates only small low-rank matrices instead of the full weight matrix.

Very parameter-efficient and storage-efficient.

Maintains strong performance while reducing computational cost.

Useful for deployment-oriented systems and repeated experimentation.

In this thesis, LoRA gave a strong balance between performance and efficiency, especially for severity classification.

## Research Contributions

- CLIP ViT-B/32 framework integrating disaster images and text
- Adapter and LoRA fine-tuning with fewer trainable parameters
- Multi-task analysis: humanitarian category, informativeness, and severity
- Explainability through Grad-CAM and token-level attribution

## Practical Applications

- Automatic analysis of disaster-related social media content
- Support for NGOs and rescue agencies in identifying urgent needs
- Rapid damage-severity assessment for faster decisions
- Interpretable predictions for AI-assisted decision systems

## Societal Benefits

- Faster disaster management through reduced manual screening
- Improved prioritization of rescue and relief resources
- Quicker identification of affected and vulnerable populations
- Greater confidence through transparent, responsible AI

## Usefulness & Future Potential

- Balances performance, efficiency, and explainability
- Real-time disaster monitoring platforms
- Humanitarian decision-support and edge-based analysis
- Future integration with emergency response networks

**“CRISISCLIP-X enables efficient, transparent, and practical multimodal AI-based disaster understanding for supporting faster and more reliable humanitarian response.”**

## Model Configuration

### BACKBONE MODEL

#### CLIP ViT-B/32

- Pre-trained vision–language model
- Image encoder + text encoder
- Image input:  $224 \times 224$
- Maximum text length: 77 tokens

### TRAINING FRAMEWORK

- PyTorch
- HuggingFace Transformers
- CUDA GPU acceleration

## Training Parameters

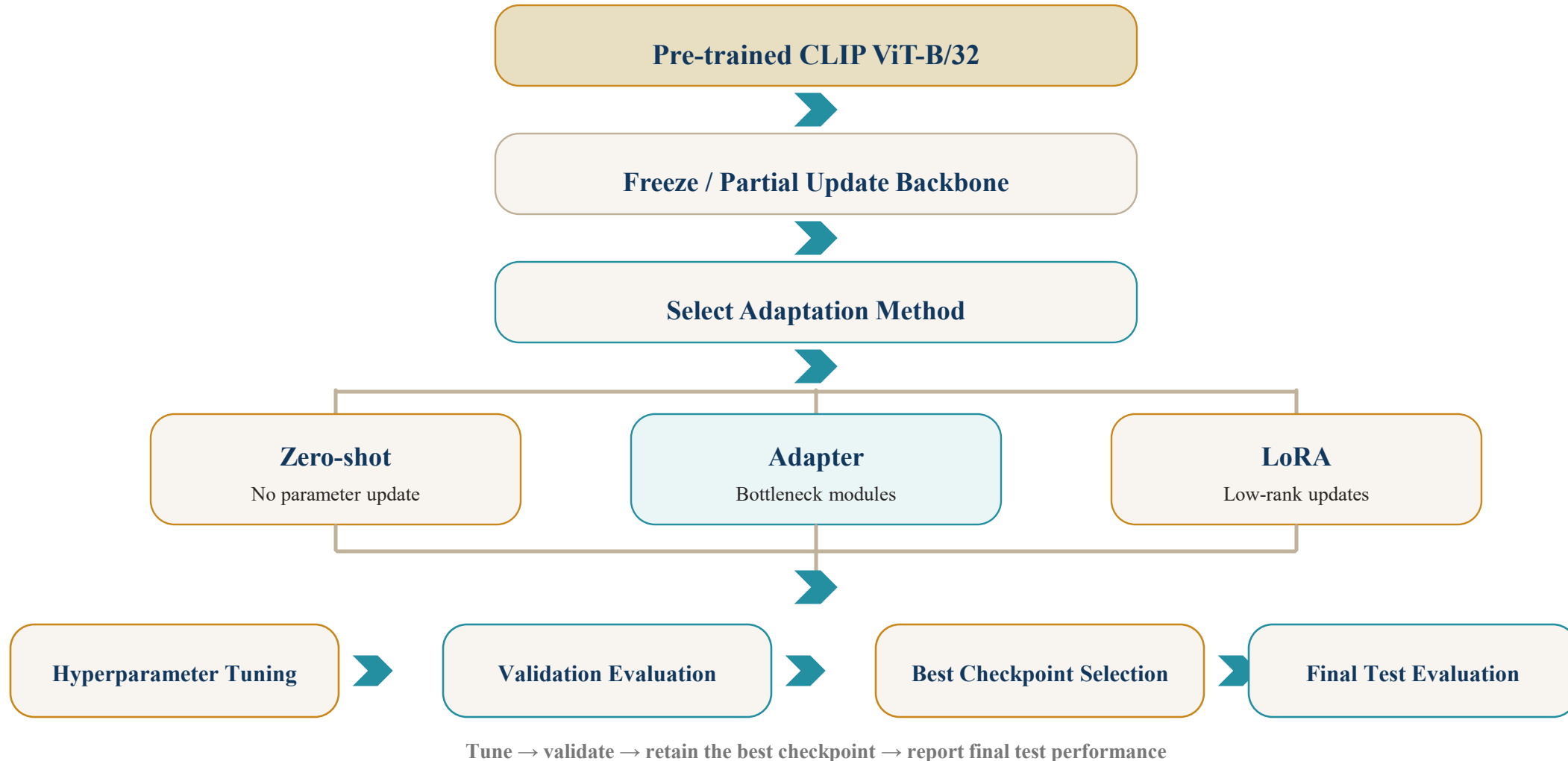
|                   |                       |
|-------------------|-----------------------|
| Optimizer         | AdamW                 |
| Weight Decay      | 0.01–0.05             |
| Loss Function     | Cross Entropy         |
| Label Smoothing   | 0.1                   |
| Scheduler         | Warmup + LR Scheduler |
| Gradient Clipping | 1.0                   |

## Fine-tuning Strategy

- 1 Zero-shot Prompting**  
No trainable parameters  
CLIP knowledge reused directly
- 2 Adapter Fine-tuning**  
Small bottleneck modules  
Lower memory; efficient adaptation
- 3 LoRA Fine-tuning**  
Low-rank weight updates  
Fewer trainable parameters

**Preserve CLIP pretrained knowledge while improving disaster classification with minimum computational cost.**

## RESEARCH WORKFLOW



## Dataset Configuration

### DATASETS

**CrisisMMD + MEDIC**

**Modalities: Image + Text**

### CLASSIFICATION TASKS

- 1 Informativeness Classification
- 2 Humanitarian Classification
- 3 Damage Severity Assessment

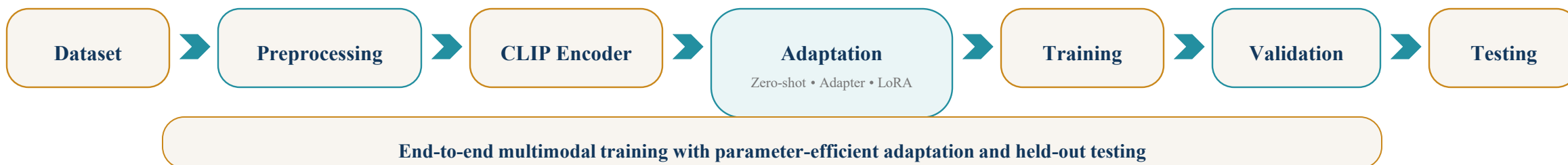
## Computational Environment

|                    |                                 |
|--------------------|---------------------------------|
| Operating System   | Microsoft Windows               |
| Python Environment | Anaconda (tweet project)        |
| CPU                | Intel Core i5-12500H (12th Gen) |
| GPU                | NVIDIA GeForce RTX 3050 Laptop  |
| GPU Memory         | 4 GB                            |
| System Memory      | 16 GB                           |
| Storage            | Samsung NVMe SSD 512 GB         |

## Implementation Framework

- 1 Deep Learning  
**PyTorch**
- 2 Multimodal  
**Hugging Face Transformers**
- 3 Efficient Fine-tuning  
**PEFT**
- 4 Deployment  
**Streamlit**

### COMPLETE WORKFLOW



*This section presents quantitative performance, visual analysis, explainability, and comparative findings across the three classification tasks.*

## Tasks Evaluated

Informativeness classification  
Humanitarian-category classification  
Damage-severity classification

## Evidence Used

Confusion matrices and ROC curves  
t-SNE visualisation and example predictions  
Grad-CAM, attention rollout, token attribution, and SHAP-style analysis

## Result Trend

Fine-tuned models consistently outperformed zero-shot baselines.  
Adapter tuning was strongest for humanitarian analysis.  
LoRA remained a strong efficiency-performance trade-off.

## Main quantitative findings

**Informativeness**  
**89.36%**

Best accuracy with partial fine-tuning

**Humanitarian**  
**90.58%**

Best accuracy with adapter fine-tuning

**Severity**  
**71.27%**

Best accuracy with adapter fine-tuning

## Informativeness

## Core Performance

Best Accuracy

**89.36%**

Partial fine-tuning

LoRA Accuracy

**87.17%**

Efficient variant

ROC-AUC

**0.9593**

Strong separation

Normalized Confusion Matrix

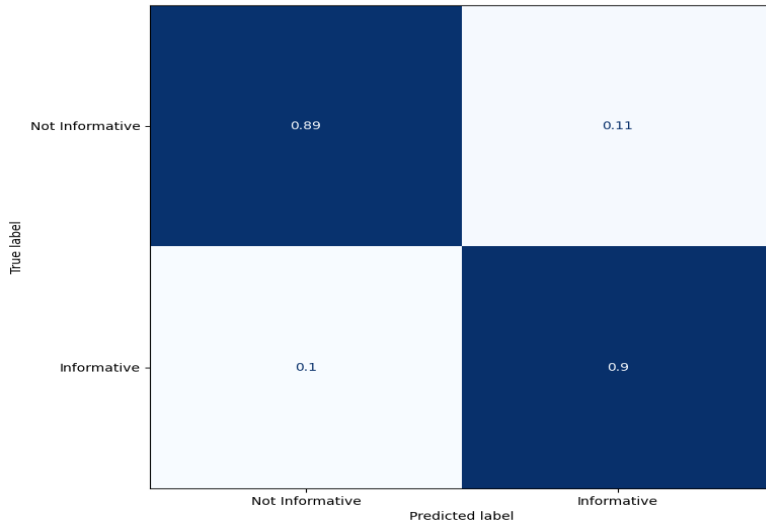


Figure : Normalized Confusion Matrix of Informativeness

Receiver Operating Characteristic (ROC) Curve

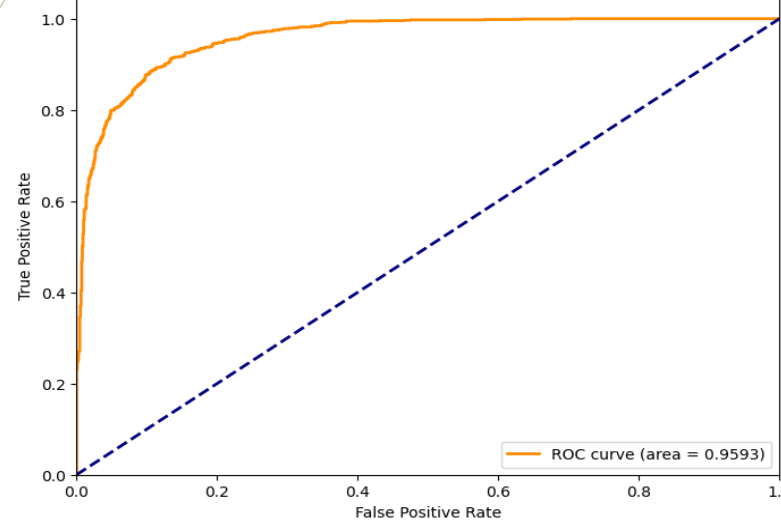


Figure : ROC Curve of Informativeness

### What this shows

Both classes show strong recall: Not Informative  $\approx 0.89$  and Informative  $\approx 0.90$ .

The high ROC-AUC indicates that the model separates informative and non-informative posts reliably.

This task works well as the first gate before humanitarian and severity analysis.

## Informativeness

## Embedding Space and Sample Predictions

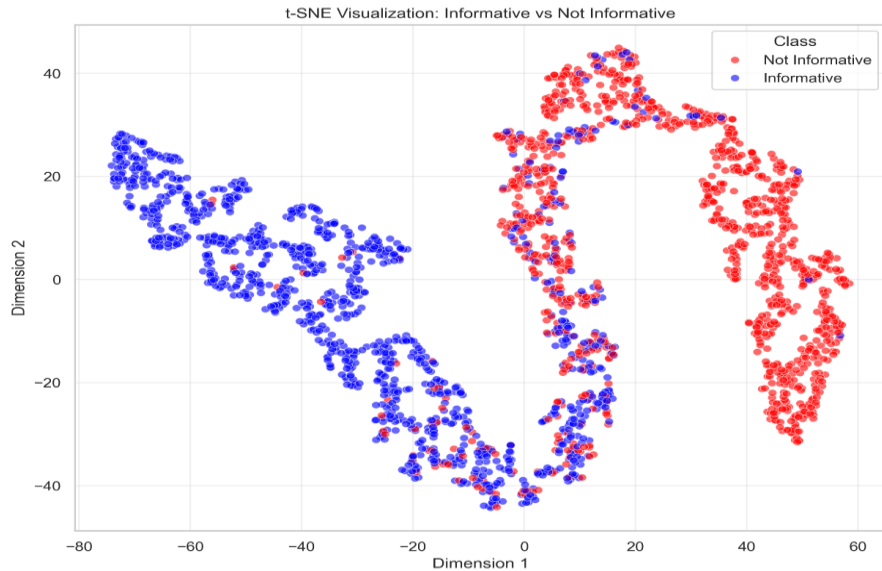


Figure:t-SNE Visualization of Informativeness

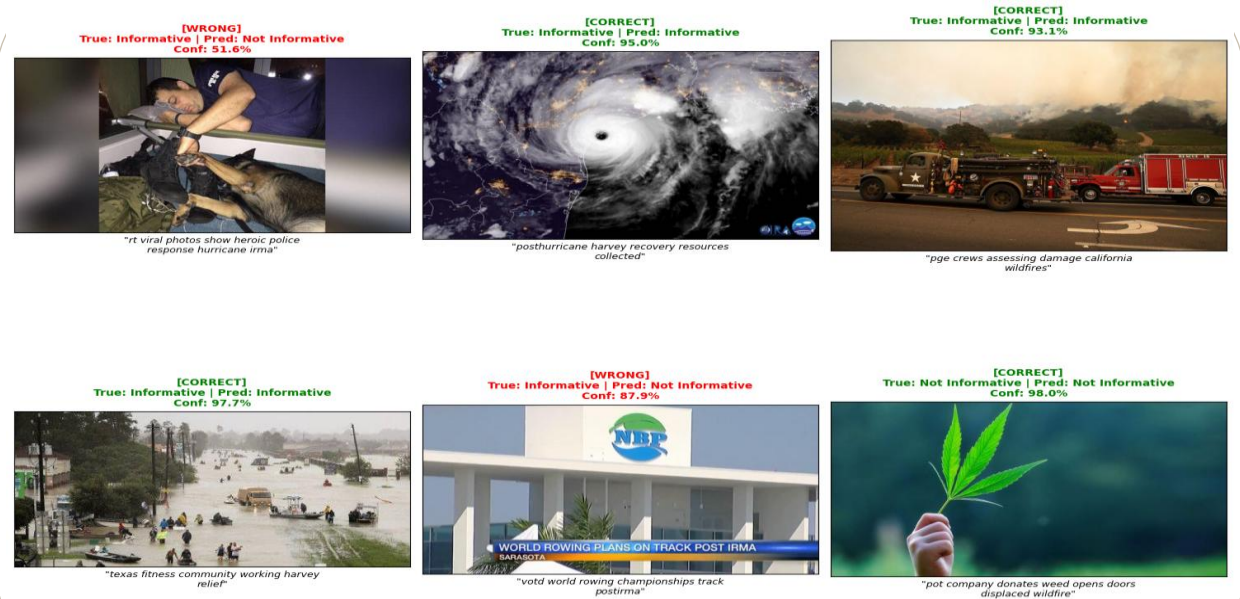


Figure : Representative Predictions of Informativeness

## Interpretation

The embedding space shows clear separation between Informative and Not Informative examples, but overlap remains for ambiguous posts. Wrong examples usually involve weak image-text alignment, misleading captions, or visually non-disaster content.

## Informativeness

## Explainability Evidence

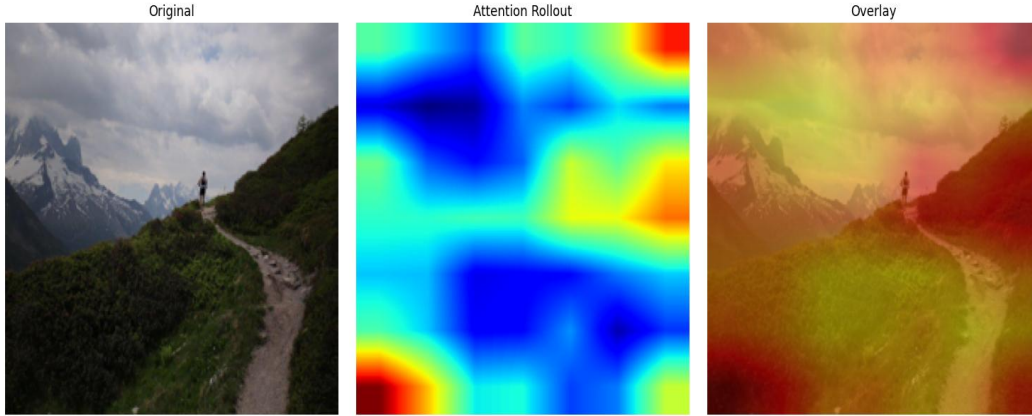


Figure: Grad-Cam Attention Rollout

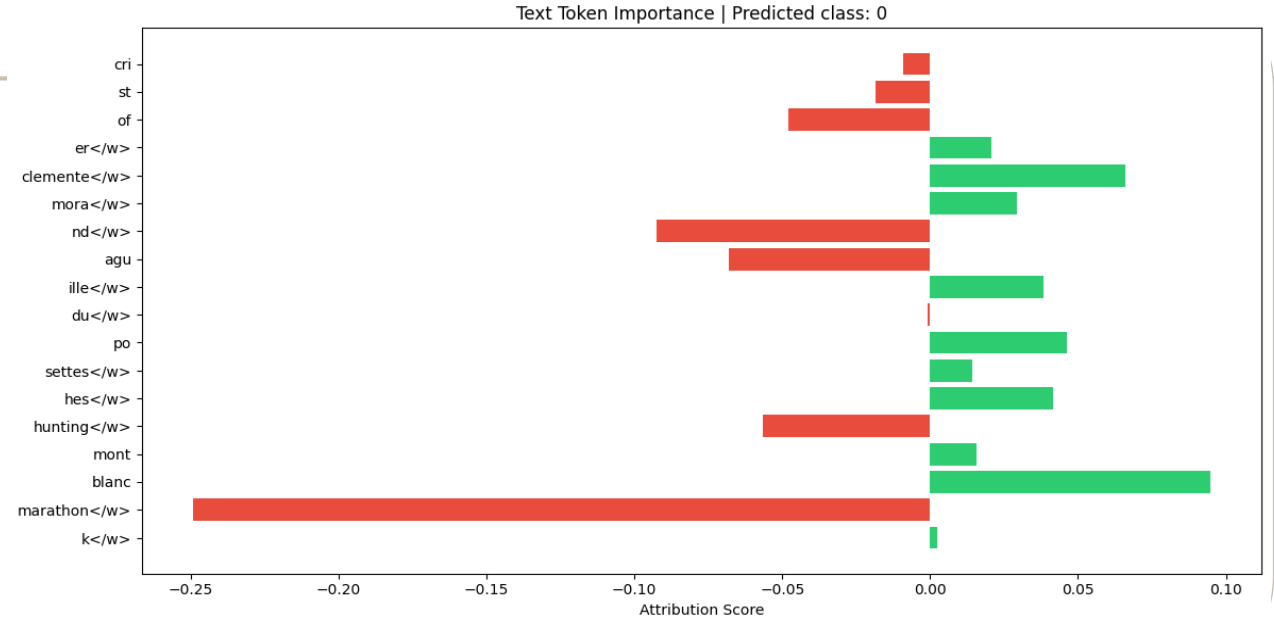


Figure :Text Token Importance of Informativeness

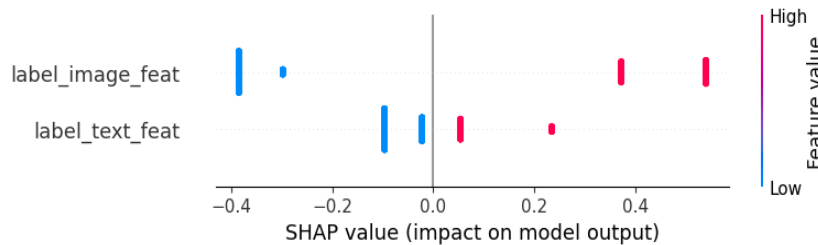


Figure : Modality-Level SHAP

### Main message

XAI confirms that both modalities contributed to the informativeness decision.

Visual attention highlights image regions that influenced the decision, while token importance shows the text cues that pushed predictions toward or away from the target class.

These explanations support human review but should not be treated as proof of correctness.

# Results & Analysis(cont'd)

## Economic Feasibility & Cost-effectiveness For Informativeness

### Performance Comparison

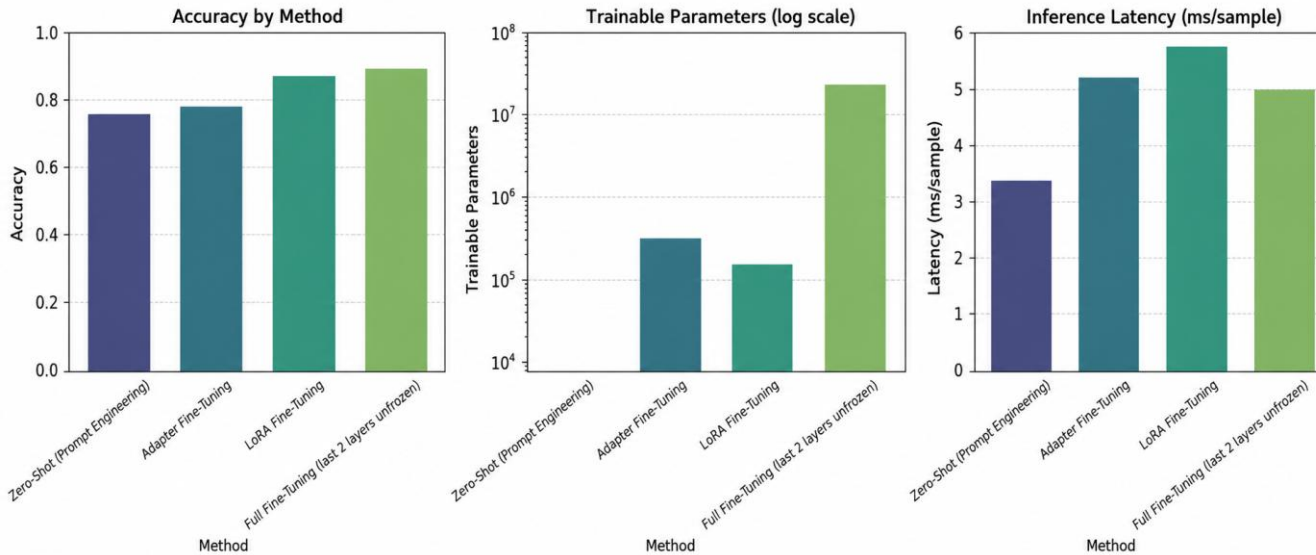


Figure : Performance Compare



Full fine-tuning achieved the highest informativeness accuracy.  
LoRA provided strong accuracy with lower trainable-parameter cost than full fine-tuning.  
Zero-shot inference was fastest and had the highest throughput.  
VRAM usage stayed almost constant across zero-shot, adapter, and LoRA.

### Computational Resource Comparison

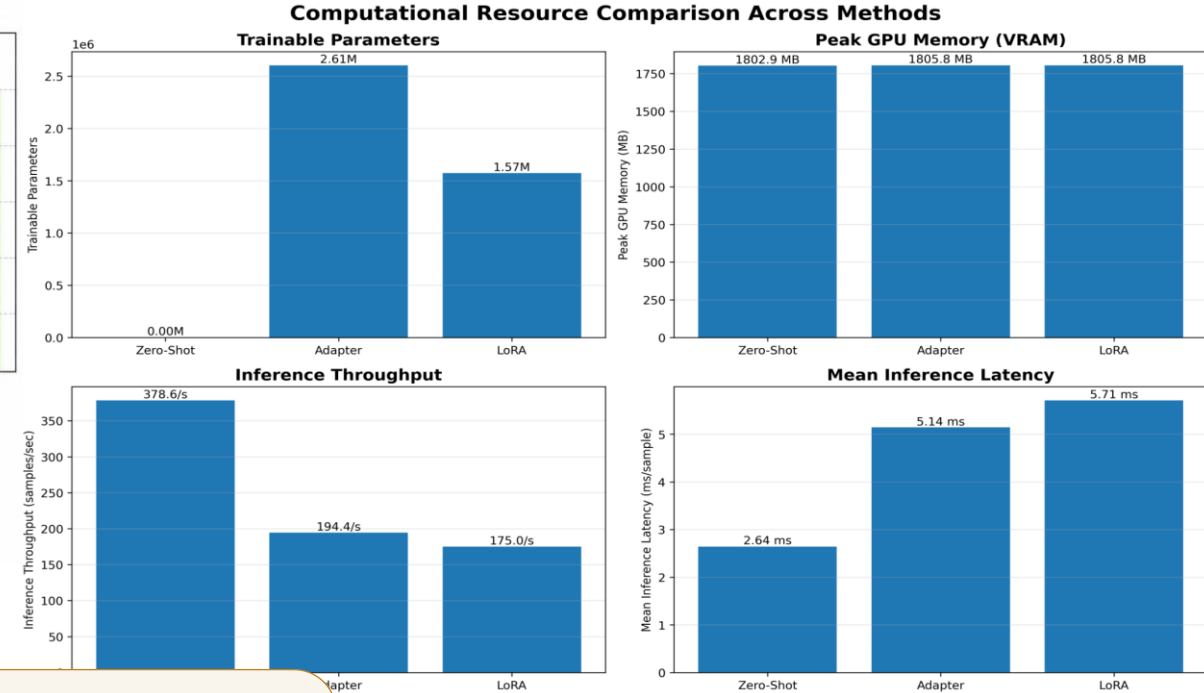


Figure : Inference count

### Result

Fine-tuning improved performance, while LoRA offered the best performance-efficiency trade-off.

## Humanitarian

## Method Comparison and Best Model

Adapter Accuracy

**90.58%**

Best model

Adapter Macro-F1

**83.24%**

Best balance

Adapter Params

**5.30M**

3.39% trainable

One-vs-Rest ROC Analysis (Humanitarian Task)

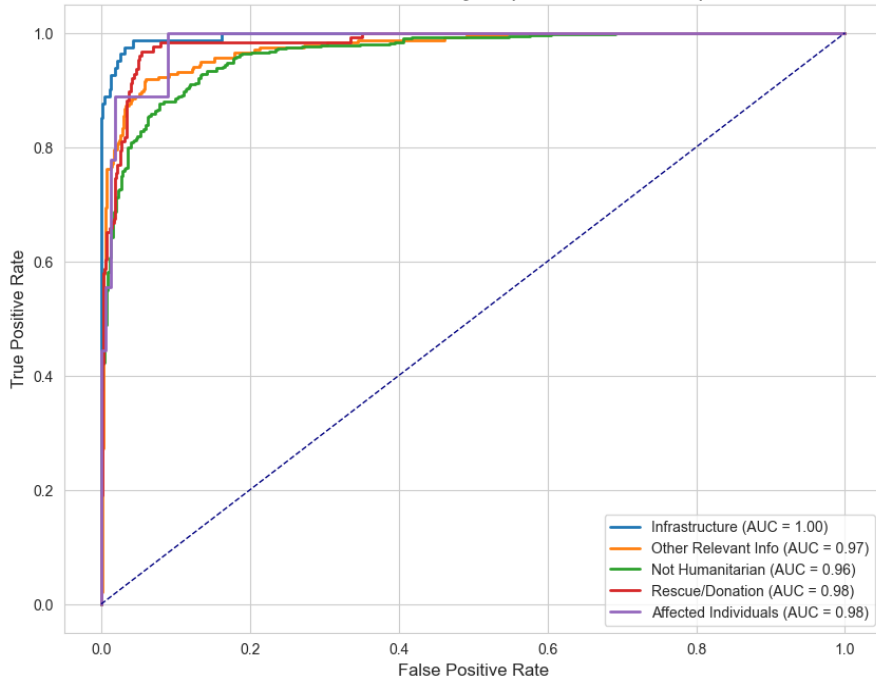


Figure : ROC curve of Humanitarian Classify

Normalized Confusion Matrix — ADAPTER

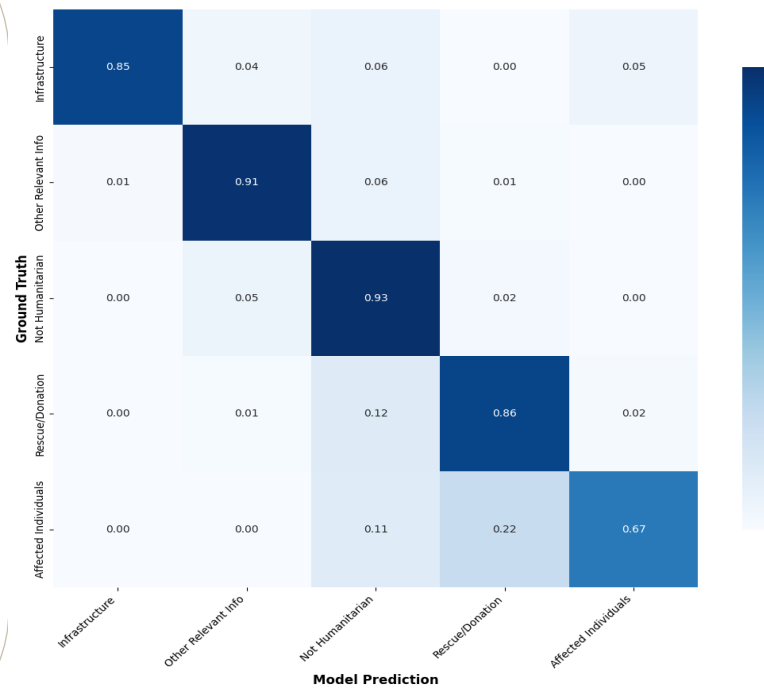


Figure : Adapter Confusion Matrix Humanitarian Classify

Normalized Confusion Matrix — LORA

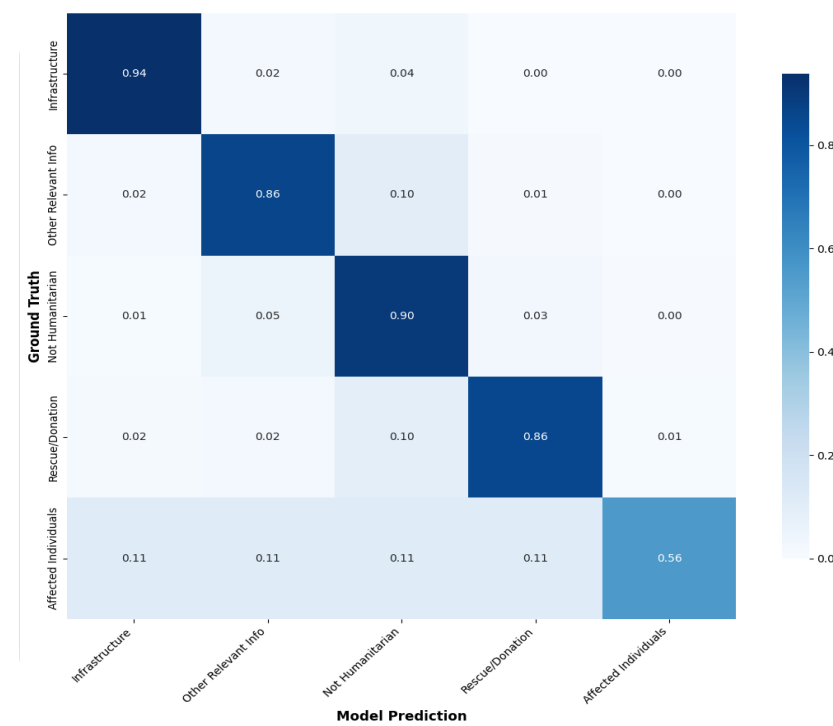


Figure : LoRA Confusion Matrix Humanitarian Classify

## Humanitarian

## Predictions and Class Separability

Qualitative Analysis: Per-Class Multimodal Predictions

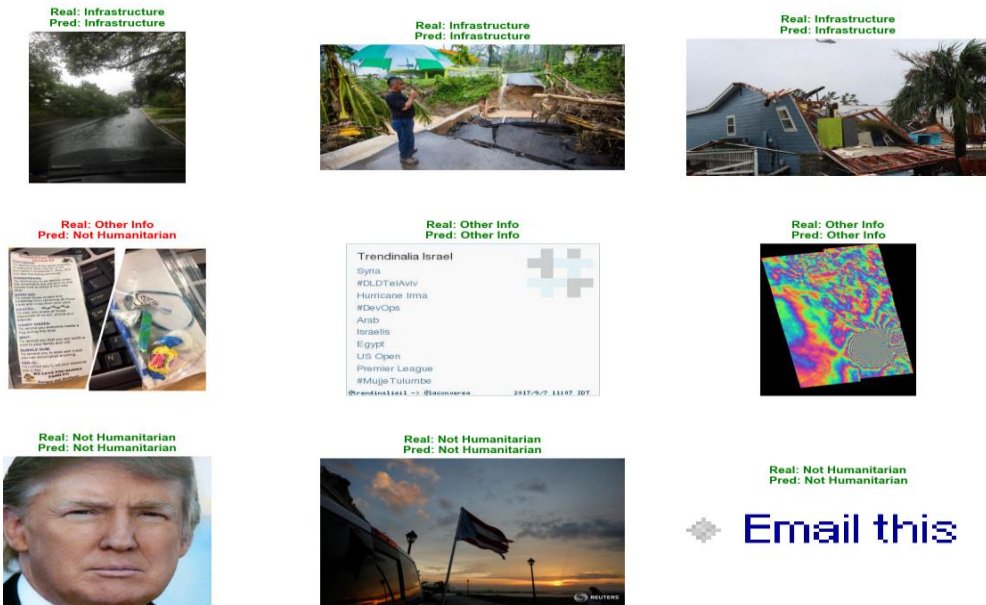


Figure : Representative Predictions Humanitarian Classify

t-SNE Projection of Fused Multimodal Embeddings

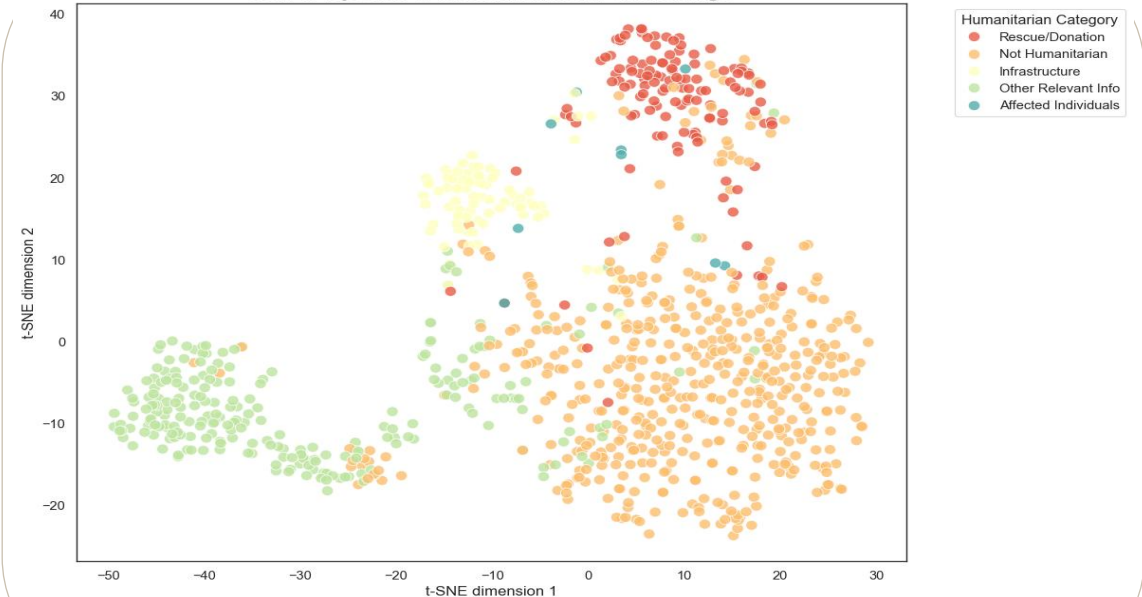


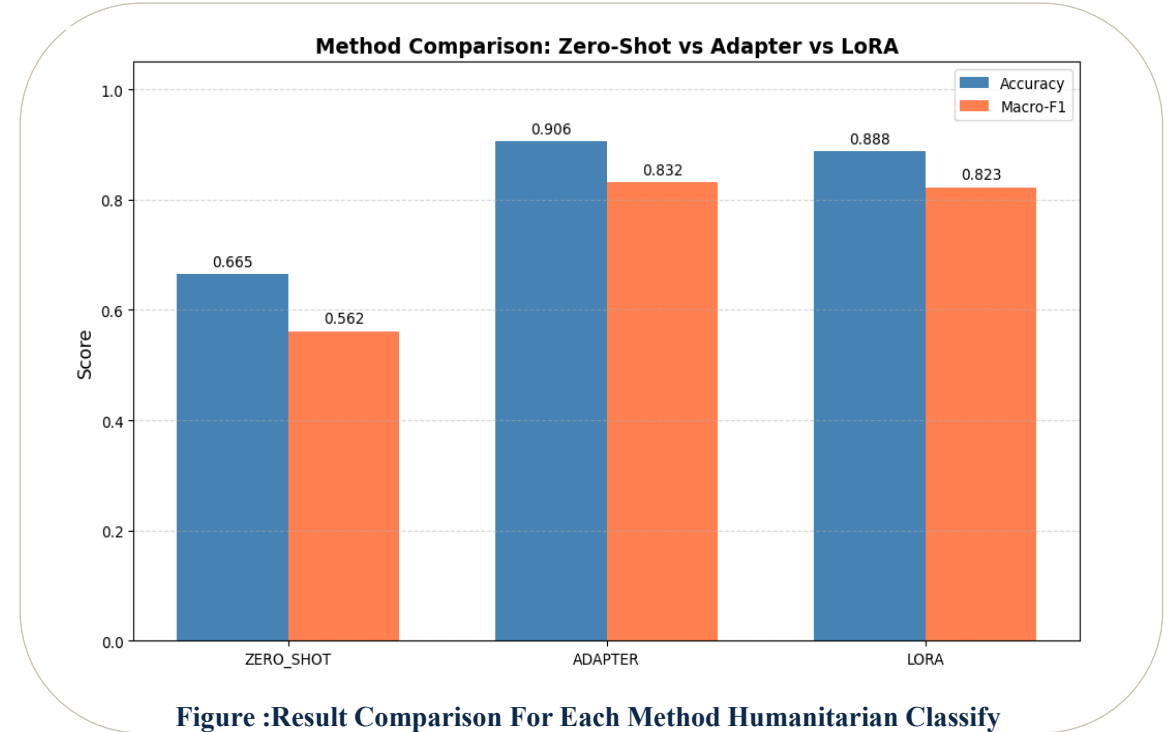
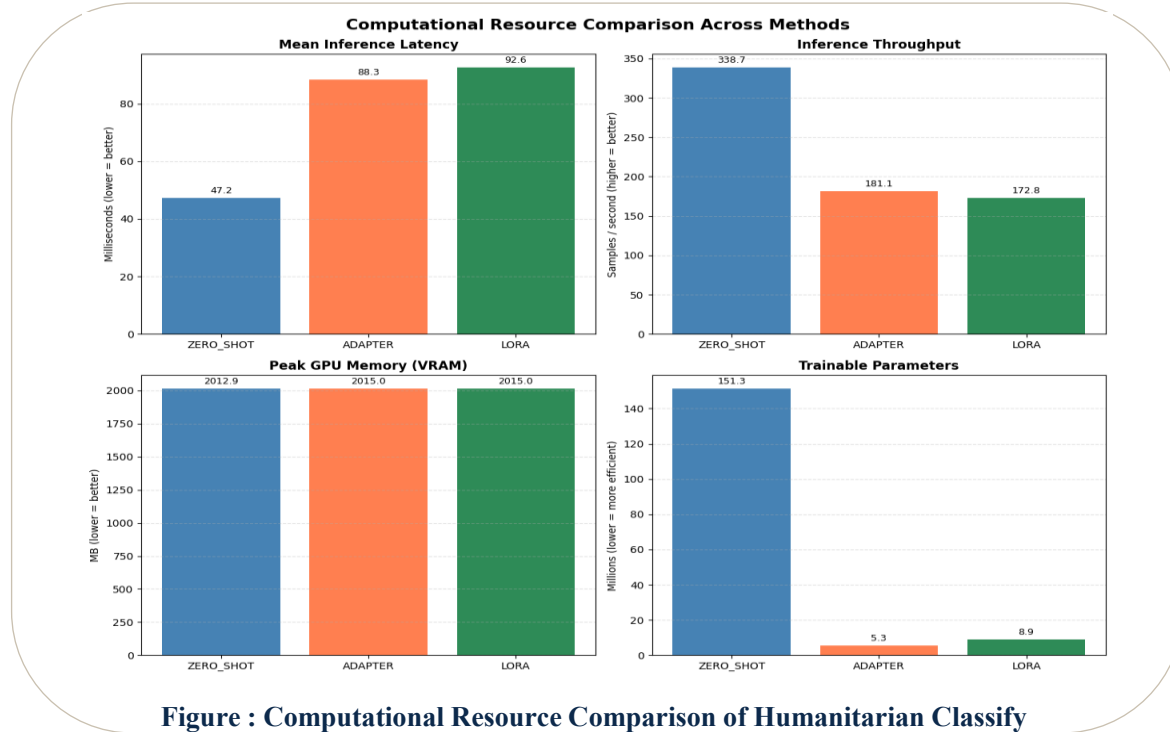
Figure :One-vs-Rest ROC Analysis Humanitarian Classify

## Why this matters

Adapter improves performance, while LoRA remains attractive for deployment because it updates fewer parameters than larger fine-tuning approaches. ROC curves show strong class separability for major humanitarian categories, supporting the reliability of the classification pipeline.

## Humanitarian

## Economic feasibility/Cost-effectiveness



### Why this matters

Adapter improves performance, while LoRA remains attractive for deployment because it updates fewer parameters than larger fine-tuning approaches. ROC curves show strong class separability for major humanitarian categories, supporting the reliability of the classification pipeline.

## Humanitarian

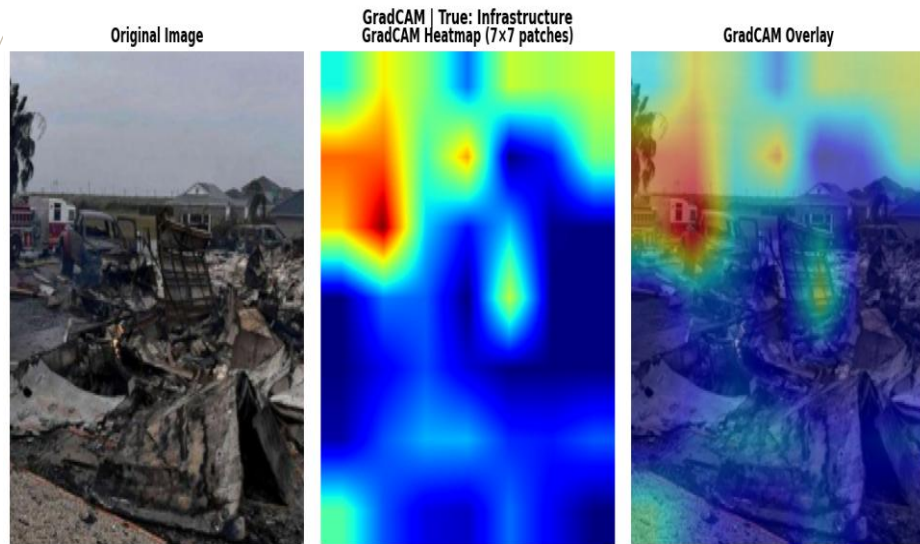


Figure : Grad-CAM Explanation Humanitarian Classify

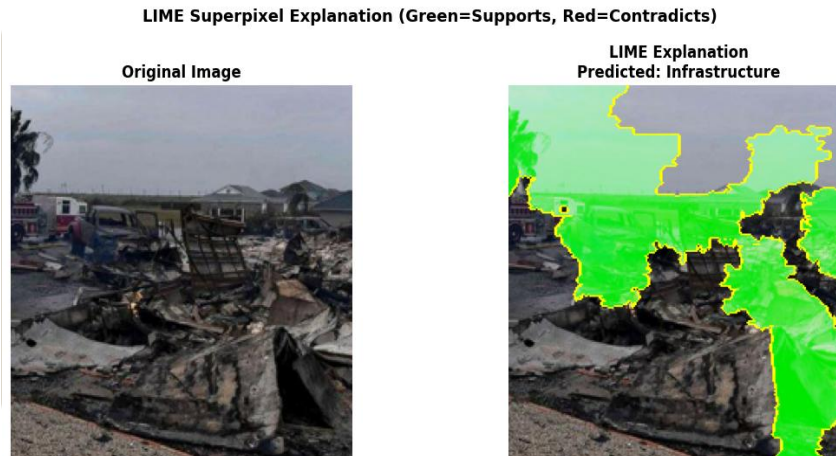


Figure : LIME Superpixel Explanation

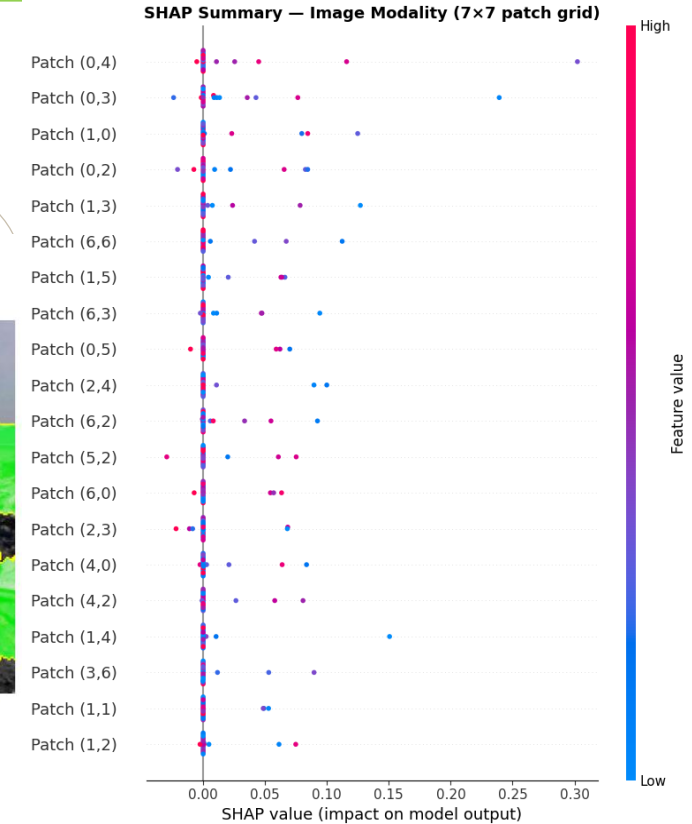


Figure : Image-Modality SHAP

Grad-CAM and LIME show that the model focused on damaged infrastructure and visually meaningful regions for humanitarian prediction.

SHAP-style image analysis indicates that certain image patches contributed more strongly to the output.

Together, these explanations make humanitarian classification more transparent for human reviewers.

## Severity

## Core Performance

Best Accuracy

**71.27%**

Adapter

LoRA Macro-F1

**58%**

Balanced classes

Weighted F1

**69%**

LoRA

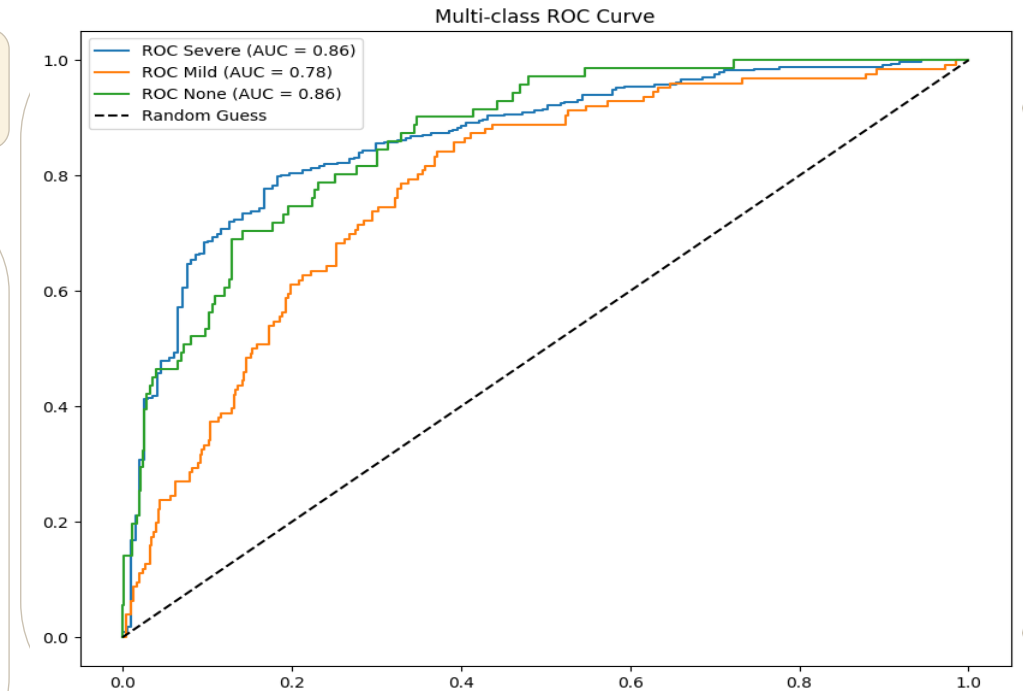
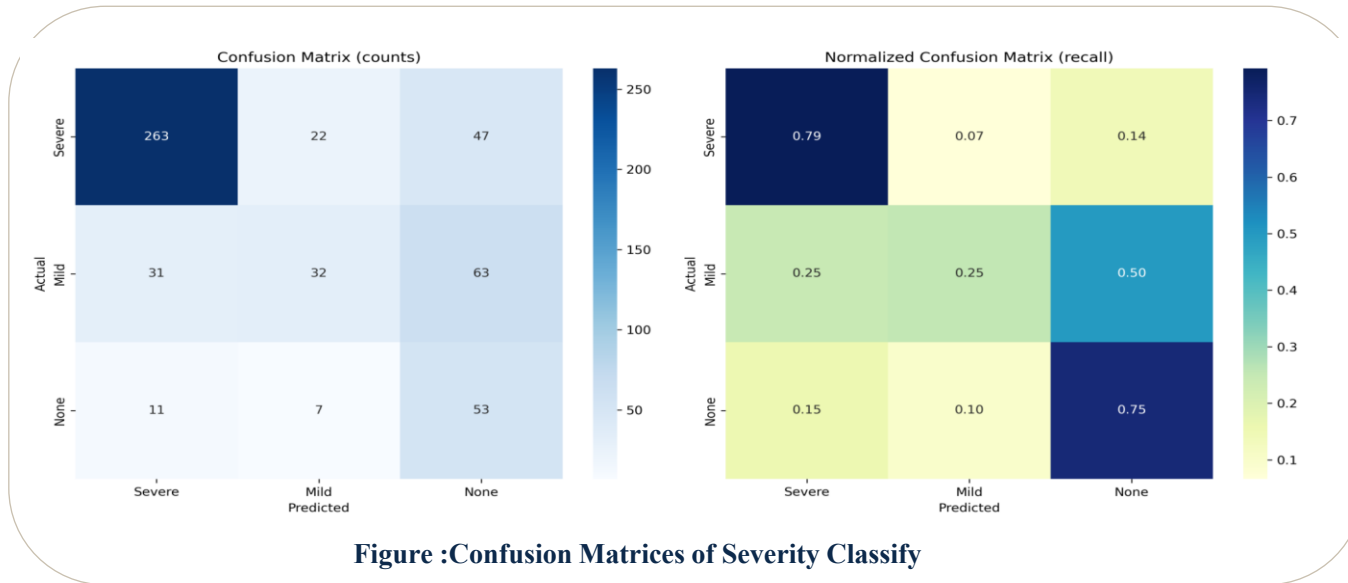


Figure : Multi-class ROC Curve of Severity Classify

Severity classification is harder because Mild damage overlaps visually with Severe and None.

ROC results show stronger discrimination for Severe and None than for Mild, which remained the most difficult severity class.

Severity

## Method Trade-off and Qualitative Predictions

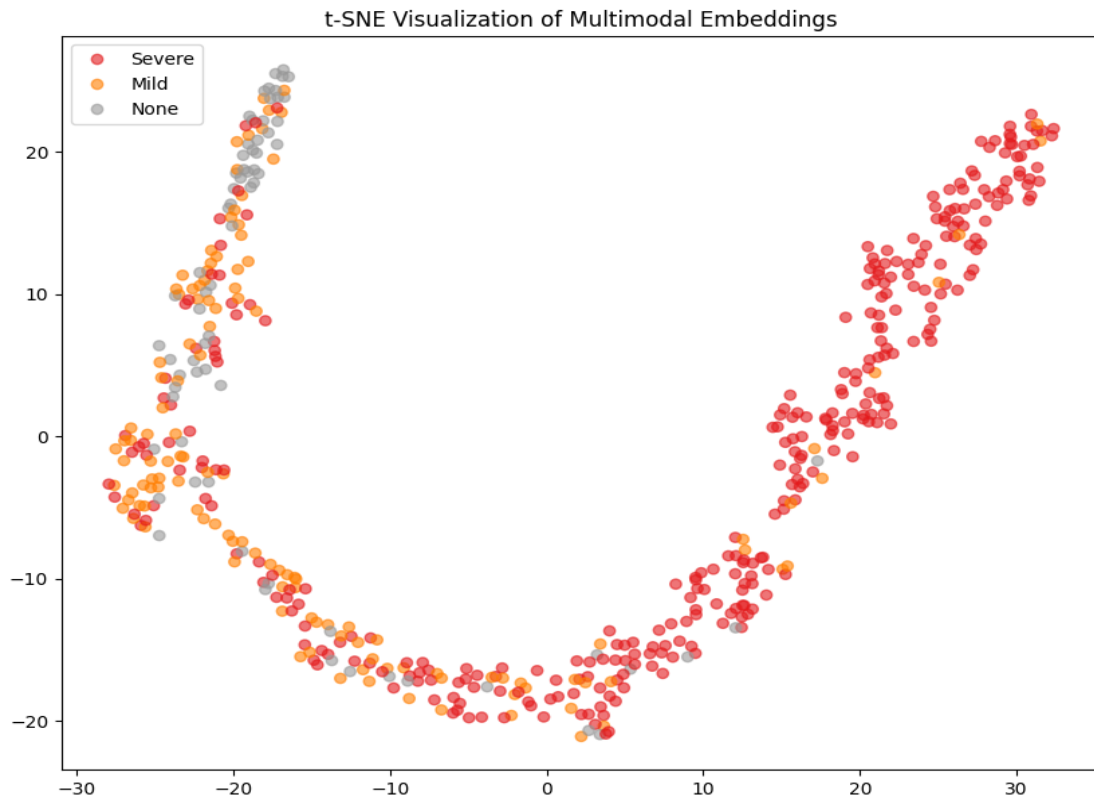


Figure:t-SNE Embedding of Severity Classify

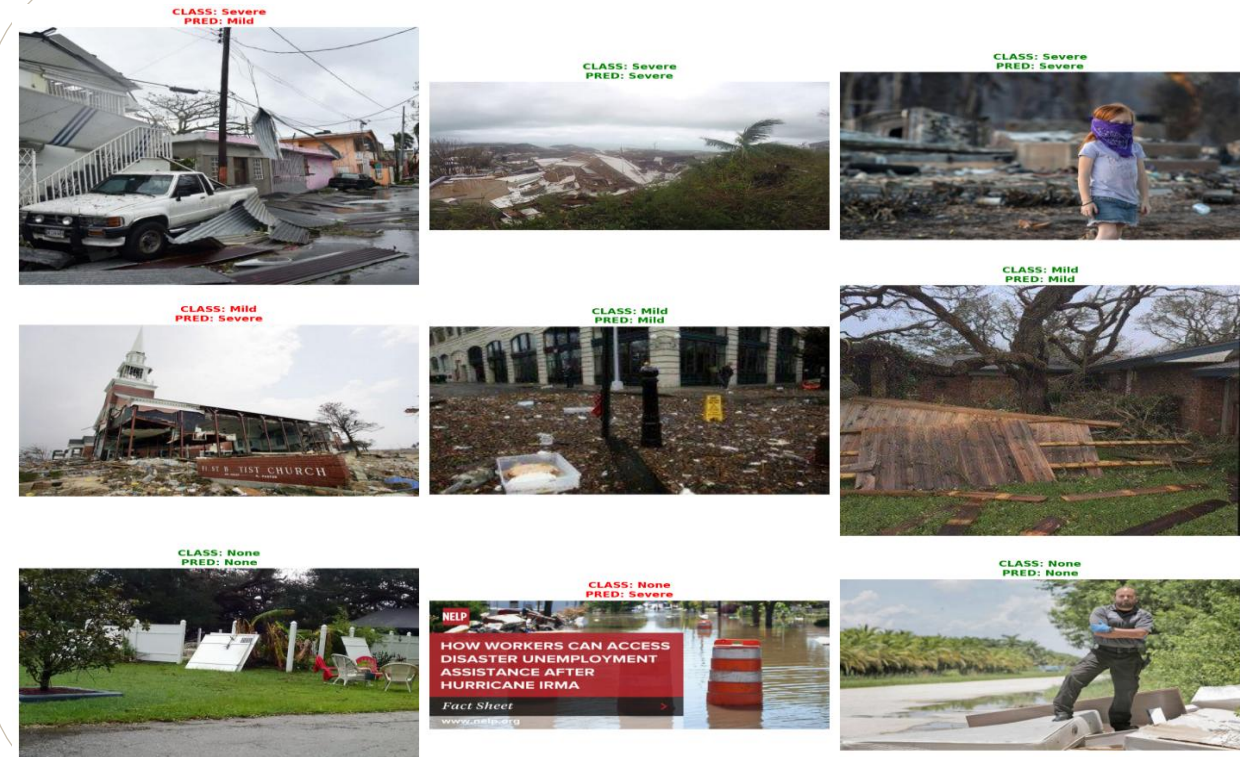


Figure :Example Predictions of Severity Classify

## Severity

## Economic Feasibility/Cost-effectiveness

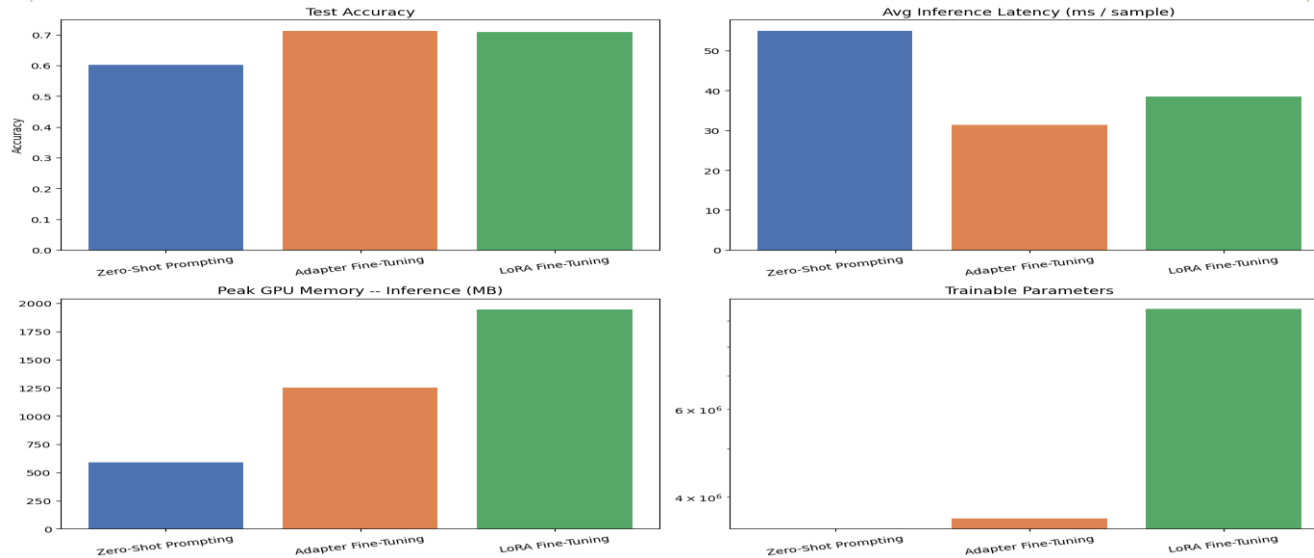


Figure :Accuracy, Latency, Memory, Parameters

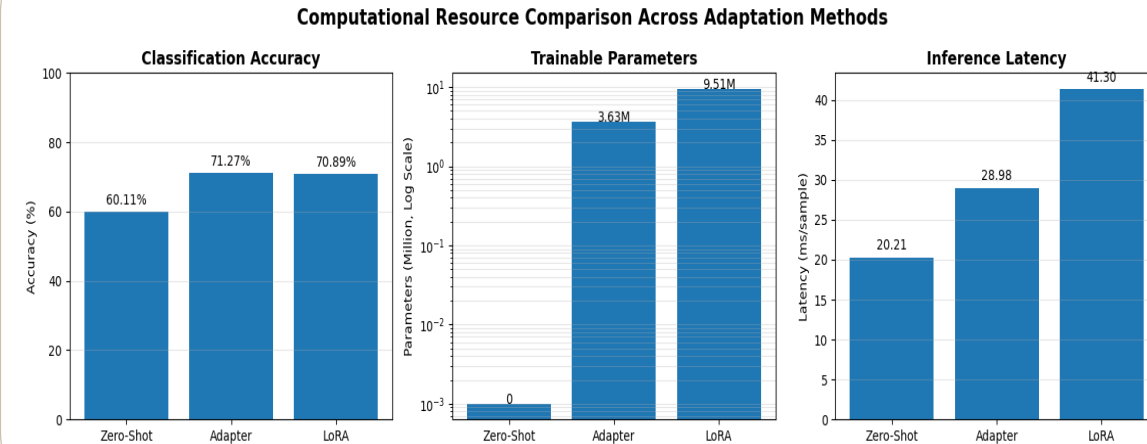


Figure :Computational Resources of Severity Classify

Adapter achieved the best accuracy, while LoRA provided a more deployment-friendly trade-off between performance and parameter efficiency. The t-SNE and examples show that clear Severe cases are easier to classify, while borderline Mild and None cases can still be confused.

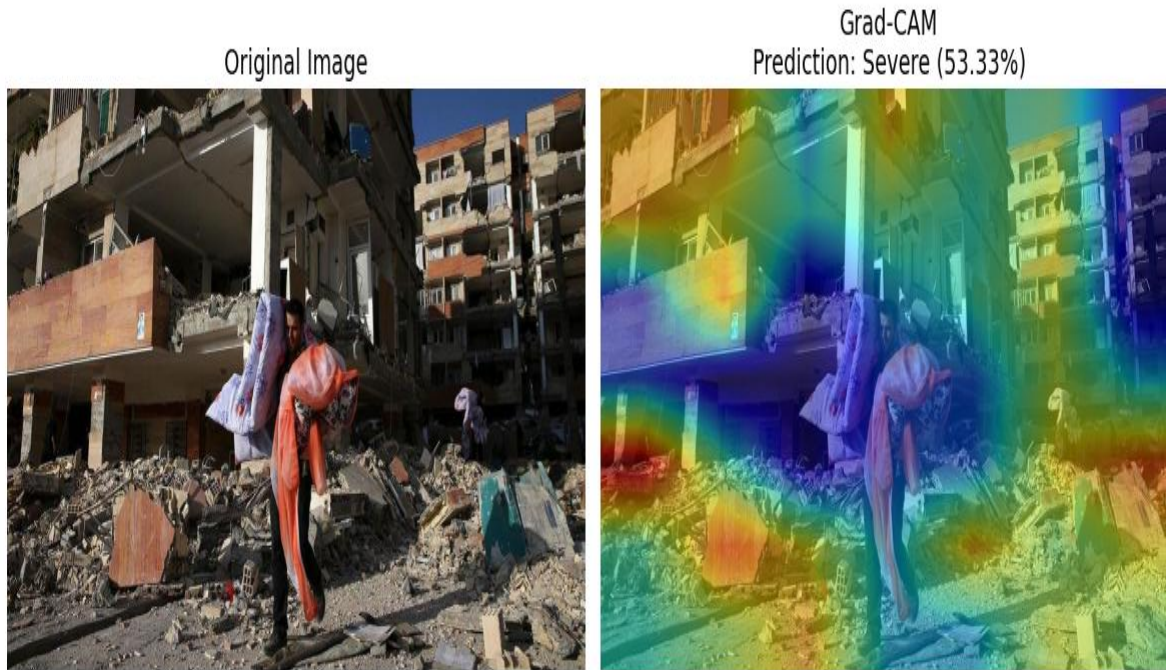


Figure: Grad-CAM for Severe Damage

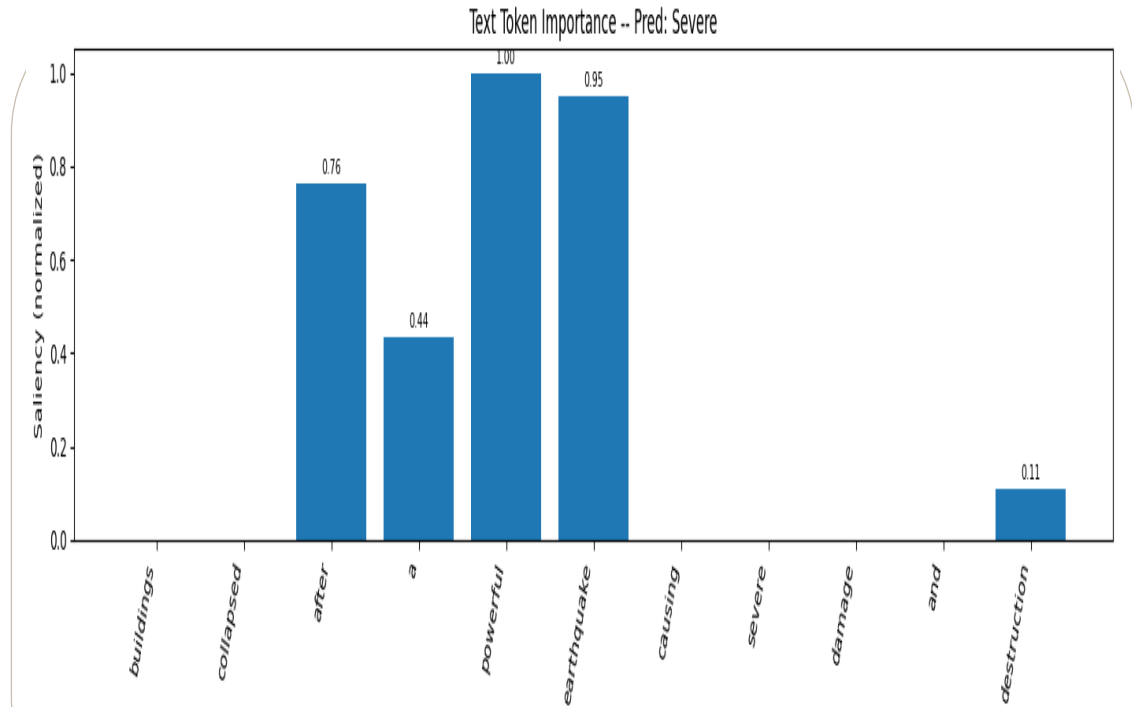


Figure :Text Token Importance of Severity Classify

## Final interpretation

Grad-CAM highlights damaged structures and debris regions that contribute to severe damage prediction. Text attribution identifies disaster-related terms such as earthquake, powerful, and destruction. The explanation indicates that CRISISCLIP-X utilizes both visual and textual disaster evidence while maintaining the need for human supervision in critical decisions.

# Results & Analysis(cont'd)

## Streamlit Web App: Why It Was Chosen

### Why Streamlit?

Fast prototype for presenting the trained model without building a complex production system.

Supports image upload and caption/tweet-text input in the same interface.

Easy to display prediction labels, confidence scores, and explanation outputs.

Suitable for thesis defense because the workflow is transparent and easy to demonstrate.

Runs locally, so the demo does not depend on external deployment or cloud hosting.

### Interface Snapshot

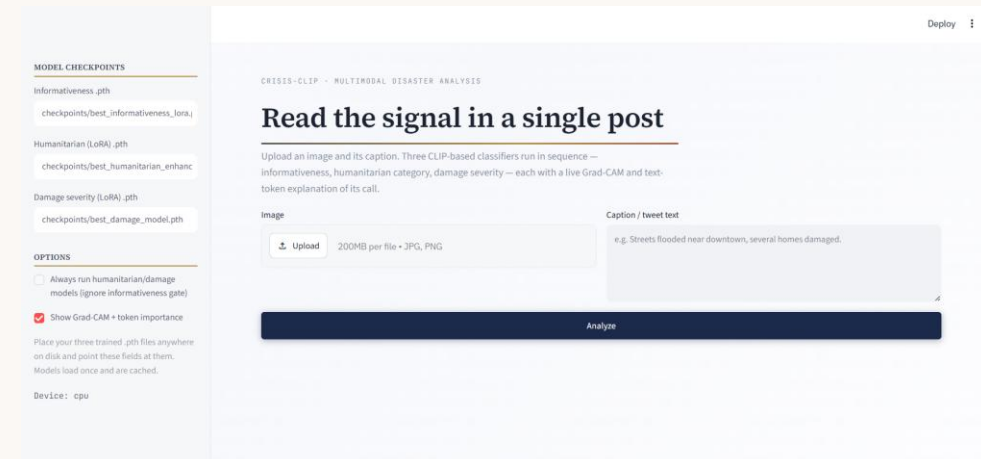
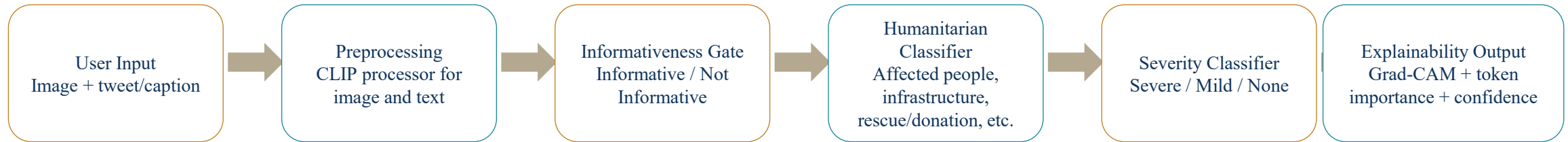


Figure : Input interface with model checkpoints, upload area, caption box, and analysis button.

**Takeaway: Streamlit was selected to convert the model into an interactive, explainable, and defense-ready prototype.**

*The application follows a sequential disaster-analysis workflow: input → gate → classification → explanation.*



## Main Components

Sidebar: paths for three model checkpoints.  
Input section: image upload and caption/tweet text area.  
Analyze button: starts the sequential prediction pipeline.  
Output section: class labels, confidence, and explanation visuals.

## Why Sequential Design?

The informativeness model works as a first filter.  
Humanitarian and severity classifiers run only when analysis is needed.  
This reduces unnecessary computation and keeps the workflow understandable.  
The design matches real disaster-response triage: screen first, then classify details.

## Streamlit Demo Workflow

MODEL CHECKPOINTS

Informativeness .pth

checkpoints/best\_informativeness\_lora.j

Humanitarian (LoRA) .pth

checkpoints/best\_humanitarian\_enhanc

Damage severity (LoRA) .pth

checkpoints/best\_damage\_model.pth

OPTIONS

☐ Always run humanitarian/damage models (ignore informativeness gate)

☒ Show Grad-CAM + token importance

Place your three trained .pth files anywhere on disk and point these fields at them. Models load once and are cached.

Device: cpu

CRISIS-CLIP • MULTIMODAL DISASTER ANALYSIS

Read the signal in a single post

Upload an image and its caption. Three CLIP-based classifiers run in sequence — informativeness, humanitarian category, damage severity — each with a live Grad-CAM and text-token explanation of its call.

Image

Upload

200MB per file • JPG, PNG

Caption / tweet text

e.g. Streets flooded near downtown, several homes damaged.

Analyze

## Sustainable AI Design

### PEFT

Adapters and LoRA update only a small share of CLIP weights, lowering training demand.

### Reuse

Pre-trained ViT-B/32 knowledge avoids costly model training from scratch.

### 4 GB

Memory-conscious experiments run on an RTX 3050 laptop GPU.

### Trade-off

Zero-shot, Adapter and LoRA comparisons support efficiency-aware model selection.

### Deploy

A lightweight Streamlit interface reduces infrastructure barriers.

## Societal & Practical Impact

### FAST

Rapidly screens disaster images and text at scale.

### PRIORITY

Highlights severity, affected conditions and humanitarian needs.

### ACTION

Turns unstructured posts into actionable information for response teams.

### TRUST

XAI explanations make predictions easier to inspect and interpret.

### USABLE

A simple interface supports use by non-technical personnel.

## Real-World Application Workflow

Social media  
disaster data



CrisisMMD +  
MEDIC  
analysis



CRISISCLIP-X  
classification



Actionable disaster  
information



Emergency planning &  
resource management

**Core contribution: efficient multimodal AI that supports faster, more transparent and accessible disaster-response decisions.**

| Evaluation Aspect                     | Existing Approaches                                                                | CRISISCLIP-X Result                                                                                  | Advantage                                                               |
|---------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| <b>Humanitarian Classification</b>    | CNN + Word2Vec + VGG16<br>F1 = 78.3%<br><br>FastText + VGG16<br>F1 $\approx$ 87.5% | <b>Adapter CLIP</b><br><b>Accuracy = 90.58%</b><br><b>Macro-F1 = 0.8324</b>                          | Improved classification using pretrained vision–language representation |
| <b>Informativeness Classification</b> | CNN + VGG16<br>F1 = 84.2%<br><br>GPT-2 + ViT-B/16<br>Accuracy = 84.66%             | <b>LoRA CLIP</b><br><b>Accuracy = 87.17%</b><br><b>Weighted F1 = 87.15%</b>                          | Higher performance with efficient parameter updates                     |
| <b>Parameter Efficiency</b>           | Conventional full-model training                                                   | <b>LoRA updates only 1.03% of trainable parameters</b>                                               | Reduced computational cost and better deployment feasibility            |
| <b>Explainability</b>                 | Most existing methods provide only predictions                                     | <b>Grad-CAM + text token attribution + XAI analysis</b>                                              | Provides interpretable visual and textual evidence                      |
| <b>Task Coverage</b>                  | Mostly a single disaster-classification task                                       | <b>Three tasks:</b><br><b>Informativeness •</b><br><b>Humanitarian Category •</b><br><b>Severity</b> | Unified multimodal disaster-analysis framework                          |

## LIMITATIONS

1

### Limited and Imbalanced Data

CrisisMMD and MEDIC cover limited disaster types, locations, languages, and time periods. Minority classes remain harder to learn.

2

### Noisy Multimodal Content

Social-media posts may contain blurred images, missing files, duplicate content, short captions, hashtags, or misinformation. Image and text may also conflict.

## FUTURE AWORK

3

### Broader Validation

Expand datasets across disaster types, regions, languages, seasons, and minority classes; validate with real field data.

4

### Robust Multimodal AI

Improve handling of missing, noisy, or conflicting modalities and test stronger fusion and XAI-faithfulness methods.

5

### Efficient Edge Deployment

Use quantisation, pruning, distillation, and hardware-aware optimisation for embedded GPU or FPGA deployment.

6

### Real-Time Response System

Integrate sensors, GPS, live dashboards, communication networks, and human supervision for responsible deployment.

**Future direction: develop CRISISCLIP-X into an efficient, explainable, human-supervised system for reliable real-time disaster response.**



Figure : Publication of this reasearch

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# Thank You

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